

# AYUSH GHOSH

U.S. Citizen

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## Education

### University of California, Berkeley

Expected May 2028

*Mechanical Engineering, B.S.; Minor in Electrical Engineering and Computer Science*

GPA: 3.95

- **Relevant Coursework:** Engineering Design and Development, Computer-Integrated Manufacturing, Three-Dimensional Modeling and Design, Mechanics, Thermodynamics, Introduction to Circuits and Devices, Electricity and Magnetism

## Work Experience

### KwikKart

May 2026 – August 2026

*Hardware Engineering Intern*

*Austin, TX*

- Re-engineered a smart tablet's rear hardware assembly in **SolidWorks**, packaging electronics and cameras into the enclosure and cutting assembly mass by **30%**; designed a spring-loaded, tool-free battery ejection mechanism
- Designed and validated forced-air cooling for an **NVIDIA Jetson AGX Orin**, iterating a custom **CNC-machined** heatsink and convection-chimney flow path with dedicated intake/exhaust fans in **ANSYS CFD** to meet tight packaging constraints
- Authored **KiCad** schematics documenting the tablet's full electronic architecture, streamlining mechanical-electrical integration and hardware debugging

### FSAE Formula Electric, UC Berkeley

January 2026 – Present

*Accumulator Mechanical Engineer*

*Berkeley, CA*

- Co-designed the team's custom **140S3P** high-voltage battery pack from concept through fabrication, driving **GD&T** tolerance stack-ups, dielectric insulation, thermal management, and **DFMA**; designed electronics packaging that integrates the cell-monitoring PCBs into the module housing
- Cut accumulator assembly mass by **18%** by re-architecting the internal high-voltage layout from **first-principles** Joule-heating and thermal-conduction analysis
- Built a test rig with an **Arduino** thermocouple array, **four-point resistance** measurement, and **Python** data logging to validate **ANSYS** coupled-field thermal simulations against measured data, root-causing hotspots that drove busbar redesigns

### Payload Container Team, Senrab Co LLC

December 2025 – May 2026

*Mechanical Engineering Intern – Signed NDA*

*San Francisco, CA*

- Developed vehicle geometry and vacuum infrastructure in **SolidWorks**, including pressure boundaries, isolation gates, and pumping subsystems for high-speed, low-pressure transit
- Built **MATLAB** physics models of payload dynamics (stability, transient loads, kinematics) to inform structural sizing and detachable shell concepts for heat flux absorption
- Qualified structural components under combined aerodynamic and thermal loading using **factors of safety** and material allowables for a high-acceleration flight environment

## Projects

### Computed Axial Lithography Printer

July 2026 – Present

*Designs for Nanomanufacturing Lab, UC Berkeley*

- Designing and building HeliCAL60, a benchtop **computed axial lithography (CAL)** 3D printer that cures full parts volumetrically by projecting tomographic light patterns into a rotating vial of photopolymer resin
- Engineering **precision motion hardware** in **SolidWorks**: mounts and structural interfaces for **50  $\mu$ m-resolution** motorized translation stages and a lead-screw drive for repeatable axial positioning
- Building a **closed-loop alignment system** that pairs a **CMOS** image sensor with the motorized stage, using real-time imaging feedback to detect and auto-correct optical/rotational misalignment

### Custom Quadcopter

June 2026 – August 2026

*Personal Project*

- Designed and validated a **~0.9 kg** quad-X airframe in **SolidWorks**, using **FEA** and first-principles calculations to size motors, propellers, ESCs, and battery to a **~3.6:1** thrust-to-weight ratio and **~12 min** flight time
- Designed a **6-layer STM32F405 KiCad** flight-controller PCB (IMU, barometer, RC receiver) and wrote firmware from scratch for sensor fusion, attitude stabilization, and motor control
- Built a tethered **flight-test fixture** and tuned **PID** gains to a stable hover within  $\pm 2^\circ$  roll/pitch under disturbances

## Skills

**CAD & Design:** SolidWorks, CATIA V5, Fusion 360, Onshape, AutoCAD, KiCad, Blender

**Analysis:** FEA, Structural Analysis, GD&T, Tolerance Stack-Ups, Thermal Analysis, CFD (ANSYS Fluent)

**Fabrication:** CNC Machining, CAM, SLA/FDM 3D Printing, Laser Cutting (FabLight), CNC Wire Bending (DIWIRE), Sheet Metal, PCB Assembly, Soldering, Rapid Prototyping

**Programming & Embedded:** C/C++, Python, MATLAB, Java, Bare-Metal Firmware, Embedded Systems

**Languages:** English, Bengali, Spanish